

A system has RAM which can store four pages simultaneously to satisfy page requests. Write a program to calculate percentage page hit and page fault if the system uses Optimal page replacement technique for the page references entered by the user i.e. 1, 2,3,4, L,2,5, !,2,3,4, 5 (No of Frames=3).

CODE:

```
GNU nano 7.2 optimal.c *
#include <stdio.h>

int main() {
    int pages[] = {1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5};
    int n = 12, frames = 3;
    int frame[3];
    int i, j, k, pos, faults = 0, hits = 0;

    for(i = 0; i < frames; i++)
        frame[i] = -1;

    for(i = 0; i < n; i++) {
        int found = 0;

        // Check hit
        for(j = 0; j < frames; j++) {
            if(frame[j] == pages[i]) {
                hits++;
                found = 1;
                break;
            }
        }

        // If fault
        if(!found) {
            int farthest = -1;
            pos = -1;

            for(j = 0; j < frames; j++) {
                int k;
                for(k = i + 1; k < n; k++) {
                    if(frame[j] == pages[k])
                        break;
                }

                if(k == n) {
                    pos = j;
                    break;
                }

                if(k > farthest) {
                    farthest = k;
                    pos = j;
                }
            }

            frame[pos] = pages[i];
            faults++;
        }
    }

    printf("Optimal Page Faults = %d\n", faults);
    printf("Optimal Page Hits = %d\n", hits);

    printf("Hit Ratio = %.2f%%\n", (float)hits/n * 100);
    printf("Fault Ratio = %.2f%%\n", (float)faults/n * 100);

    return 0;
}
```

OUTPUT:

```
laxmi@laxmi:~$ nano optimal.c
laxmi@laxmi:~$ gcc optimal.c -o optimal
laxmi@laxmi:~$ ./optimal
Optimal Page Faults = 7
Optimal Page Hits = 5
Hit Ratio = 41.67%
Fault Ratio = 58.33%
laxmi@laxmi:~$
```